

38. DEVELOPMENT OF THE PANCREAS

DURATION : 02:56

STUDY SHEET

COURSE SUMMARY

In this video on pancreatic development, you will learn how this crucial organ forms from the hepatic development phase and gastric movements. Through a detailed approach, the content covers the transitions between the anterior and dorsal pancreas, as well as the exocrine and endocrine types of pancreas. The interactions between the pancreas, liver, and other embryological structures such as the stomach and heart are explored, illustrating the complexity of these relationships and their impact on metabolic health via insulin. This lesson includes theoretical and practical elements, ideal for improving your understanding of osteopathy and embryology.

PANCREAS DEVELOPMENT

In this phase of **hepatic development** and the establishment of **gastric movement** as well as the **lesser sac**, we initially observe an **anterior pancreas** and a **dorsal pancreas**.

During the **rotation** phase, the anterior pancreas performs a large three-quarter movement, joining the posterior part to form the two types of pancreas: an **exocrine pancreas** and an **endocrine pancreas**. The exocrine pancreas secretes substances related to the system of **gastric juices** and **pancreatic juices**, such as **amylase**, **lipase**, and **protease**.

The pancreas is related to the liver, while the dorsal pancreas performs a movement that illustrates hepatic rotation. This rotation occurs simultaneously with an axis that positions itself very posteriorly in a mesentery called the **posterior mesentery**.

If you have understood the developmental movement of the lesser sac, you are beginning to grasp that everything else is merely a dynamic of this development on a transverse plane. The pancreas plays a crucial role in releasing **insulin**, which opens cellular gates, thereby influencing **mitochondrial activity** and sugar metabolism.

During dissection, it is important to note that it is very difficult to separate the **duodenum** from the pancreas, as they are deeply adhered. If one observes the structure from above, one can see a hepatic movement that rotates the stomach and creates the lesser sac, orienting itself in a specific way.

This process is comparable to a **tai chi** movement, occurring simultaneously at the level of the brain and hands. We observe the opening of the **lungs**, the **looping of the heart**, the development of the **brain**, as well as the spiral of the **diaphragmatic system**.

The anti-clockwise development of the **colon** and the expression of the **upper limbs** also occur in synchronisation. During brain flexion, it is interesting to observe the interactions at the level of the heart, the anlage of the eyes, the configuration of the face, palate, and lungs, and to assess where one is in this process.

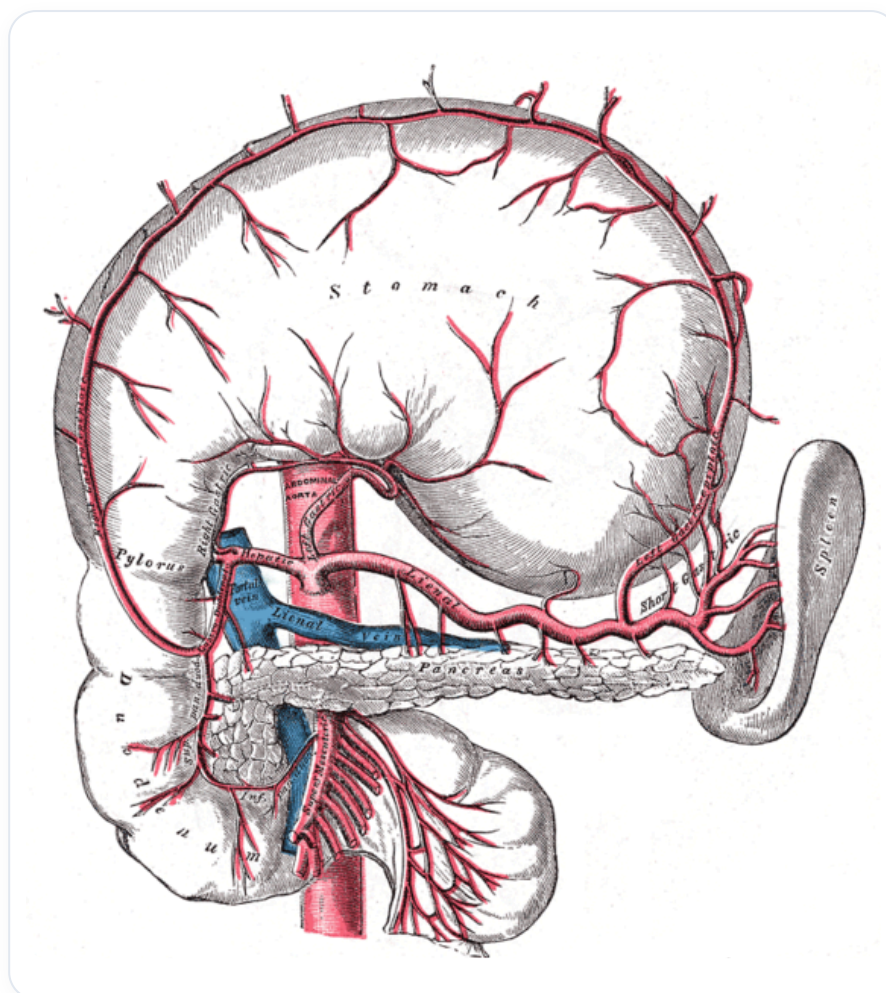


FIG. — Stomach/pancreas/spleen irrigation

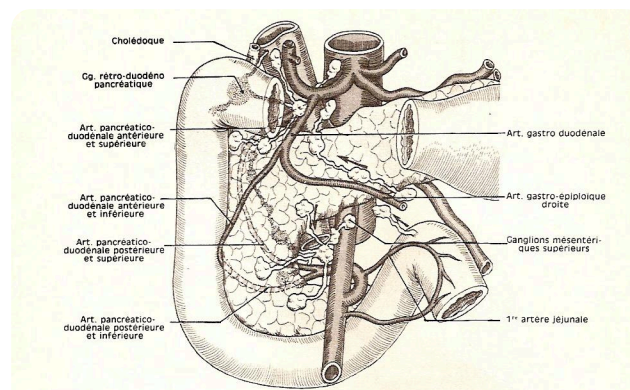


FIG. 77 b. — *Vaisseaux duodéno-pancréatiques. Vue antérieure.*

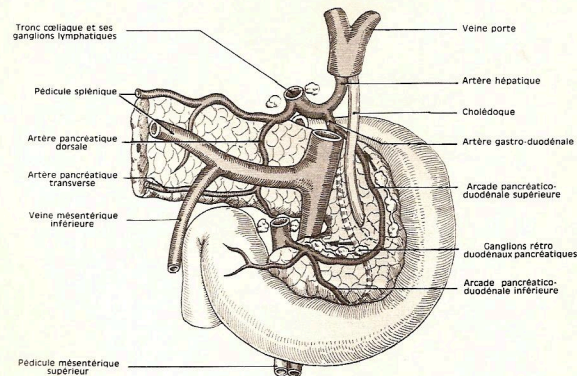


FIG. 77 c. — *Vaisseaux duodéno-pancréatiques. Vue postérieure.*

FIG. — *Duodenopancreatic vessels*